

# KARLA VASQUEZ

831-789-0338 | [kdvasque@gmail.com](mailto:kdvasque@gmail.com) | [kdvasquez.github.io](https://kdvasquez.github.io) | [linkedin.com/in/kdvasquez](https://linkedin.com/in/kdvasquez)

## SKILLS

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- **Languages:** TypeScript, JavaScript, Python, C/C++, Swift, Java, SQL, HTML/CSS
- **Frameworks & Tools:** Angular, Tailwind CSS, SwiftUI, Astro, Firebase, REST APIs, Git, Unix, MongoDB
- **Spoken Languages:** English (Native), Spanish (Native)

## EDUCATION

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### University of Southern California

Aug 2021 - May 2025

B.S. Computer Engineering and Computer Science

Los Angeles, CA

- **Relevant Coursework:** Data Structures & Object Oriented Design, Embedded Systems, Algorithms, iOS App Development, Full Stack Web Development, IoT, Computer Networks
- **Involvement:** Autonomous Networks Research Group, Association for Computing Machinery, Creative Labs, Women in Engineering, Viterbi Impact Program

## WORK EXPERIENCE

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### Software Engineer Intern

Sept 2025 - Present

Platia

Remote

- Migrated and re-architected a production financial admin dashboard from **Angular 16** to **Angular 19**, reaching full feature parity on an **esbuild** build pipeline (**TypeScript** 4.9 to 5.7)
- Cut runtime dependencies **73%** (49 to 13) and components **~60%** (81 to 31) by consolidating four overlapping CSS frameworks into a **Tailwind CSS** and **Angular Material** design system
- Rebuilt **authentication**, user management, and transactional workflows against **REST APIs** with standardized HTTP services and error handling

### Academic Tutor (ELA & Math)

Sept 2025 - Present

HeyTutor

Soledad, CA

- Tutor **15+ students** across grades **3–5** in English and Math, leading groups of **5** in **45-minute** sessions and adapting pacing to **mixed learning levels** in real time
- Track student growth with structured beginning, midpoint, and final assessments to identify gaps and measure improvement

### Undergraduate Researcher

May 2023 - Aug 2023

USC ANRG Summer Research Academy

Los Angeles, CA

- Conducted research on AI-driven personalized learning using **GPT-3.5**; authored a **sole-author abstract** accepted to **IFoRE 2023 Research Forum**
- Analyzed weekly user-interaction data to identify engagement patterns, iterating research methodology with a faculty advisor

## PROJECTS

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### MindSpark (Swift, SwiftUI, Anthropic Claude API)

April 2026

- Built an AI-powered iOS learning app with two practice modes, integrating the **Claude API** across **5 features** — adaptive hints, story generation, wrong-answer explanations, difficulty tuning, and progress summaries — for elementary students
- Designed a **protocol-based AI service** with mock and live implementations enabling simulator-free testing, backed by **async/await** networking, **UserDefaults** + **Codable** persistence, and **XCTest/XCUI**Test coverage

### Personal Portfolio Website (Astro, TypeScript, SCSS)

March 2026

- Customized an **Astro** template into a personal portfolio with content collections, an **MDX** blog, a category-filtered project gallery, and project case-study pages

### WanderPost (Swift, SwiftUI, Firebase)

May 2025

- Developed an **MVVM** iOS postcard journal that turns photos into shareable postcards with **MapKit** pins, **CoreLocation** tagging, and live weather from the **OpenWeatherMap REST API**
- Engineered on-device persistence with **Codable** that works fully offline, with image and metadata upload to **Firebase Storage** and **Firestore** for cloud sharing, plus UI-surfaced error handling

### The SCAvenger (C, Embedded Systems)

Feb 2025 - April 2025

- Designed an environment-aware puzzle game with GPS-triggered challenges, achieving **60% RAM reduction** on a **2KB SRAM microcontroller** while integrating **4 I2C sensors** within a **\$100 budget**